Routes to Revisit Later

Numerical, Brute-Force, and Heavier Search Strategies

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This file collects routes that are potentially useful but currently less appealing for the main narrative: heavy interval searches, brute-force finite-dimensional exclusions, numerical local constants, and conditional band-limited closure arguments. They are separated from `convolution-equations.tex` so that the main file keeps a cleaner analytic progression.

1 Local Numerical Constants for the Gap

The main file formulates the local isolation radius abstractly in terms of $\underline{\nu}$ and M_0. The paragraphs below record the concrete Bessel-integral computations and numerical targets that may be revisited when one wants an explicit value of the gap.

The finite-mode numerical checks below refer to the target

$$\nu_0 \geq 0.4\,\Lambda_0. \tag{1}$$

Current certified finite-mode check. The script `numerics/circle_gap_flint_verify.py` gives a first ball-arithmetic certificate for the small-mode part of this target. It uses `python-flint` Arb balls, evaluates the Bessel functions at the midpoints of the radial

mesh, and encloses the variation on each interval by the elementary bounds $|J_m| \le 1$ and $|J_m'| \le 1$ for integer m on the real line. With cutoff $R = 200$ and the conditional tail majorant $|J_m(r)| \le (2/(\pi r))^{1/2}$ on $[R, \infty)$, the current runs prove

$$\frac{\mathcal{J}_2}{\mathcal{J}_0} \le 0.117364, \qquad \frac{\mathcal{J}_3}{\mathcal{J}_0} \le 0.091870, \qquad \frac{\mathcal{J}_n}{\mathcal{J}_0} \le 0.072366 \quad (4 \le n \le 12).$$

The first number is the bottleneck and was obtained with mesh $h = 0.0025$; the other displayed bounds use $h = 0.01$. In particular, for $2 \le n \le 12$ the finite-mode certificate proves

$$\frac{\mathcal{J}_n}{\mathcal{J}_0} < 0.12,$$

and hence

$$|5\mathcal{I}_n - \Lambda_0| > 0.4\Lambda_0, \qquad |\mathcal{I}_n - \Lambda_0| > 0.4\Lambda_0 \qquad (2 \le n \le 12).$$

The tail majorant is a theorem (not conditional): $|J_0(r)| \le \sqrt{2/(\pi r)}$ for all $r > 0$ (Watson §7.31), and $|J_m(r)| \le \sqrt{2/(\pi r)}$ for $r \ge m \ge 1$ (Krasikov 2006). Combined with the explicit large-mode bound in Lemma 2.13, the target (1) is now fully proved: Theorem 2.16 certifies $\nu_0 \ge 0.4\Lambda_0$.

The Lipschitz constant M_0 via sixfold Bessel products. The second derivative $D^2T(f_0)$ of $T(f) = E^*(|Ef|^4 Ef)$ at the constant decomposes in Fourier modes. Using the identity $D^2N(u_0)[v_1, v_2] = u_0^3[6v_1v_2 + 6v_1\bar{v}_2 + 6v_2\bar{v}_1 + 2\bar{v}_1\bar{v}_2]$ (which follows from differentiating $N(u) = |u|^4 u$ at a real point u_0), each input mode pair $(e^{in_1\theta}, e^{in_2\theta})$ produces output at angular modes $\{n_1 \pm n_2, -(n_1 \pm n_2)\}$, with radial coefficients proportional to the sixfold Bessel integrals

$$\mathcal{B}(a, b, c) := \int_0^\infty J_0(r)^3 \, J_a(r) \, J_b(r) \, J_c(r) \, r \, dr.$$

The Hilbert–Schmidt norm of $D^2T(f_0)$ as a bilinear form $L^2(S^1) \times L^2(S^1) \to L^2(S^1)$ is

$$\left\| D^2T(f_0) \right\|_{\mathrm{HS}}^2 = (2\pi)^6 \sum_{n_1, n_2 \in \mathbb{Z}} \sum_k |S_k(n_1, n_2)|^2 \, |\mathcal{B}(|n_1|, |n_2|, |k|)|^2 ,$$

where S_k collects the phase-weighted angular coefficients at mode k. By the inequality $\|D^2T\|_{\mathrm{bil}} \le \|D^2T\|_{\mathrm{HS}}$, this gives an upper bound on the Lipschitz constant of DT in a ball around f_0. Accounting for the eigenvalue variation (Λ is coupled to f via $\Lambda = \langle T(f), f \rangle$) and the sphere projection, a safe upper bound for the full Lipschitz constant of the sliced Euler–Lagrange derivative is $M_0 \le 2 \, \|D^2T\|_{\mathrm{HS}}$.

A floating-point evaluation (script `numerics/newton_kantorovich.py`, $R = 2000$, $h = 0.05$, modes $|n_j| \le 6$) gives

$$\left\| D^2T(f_0) \right\|_{\mathrm{HS}} \approx 1831, \qquad M_0 \le 3662, \qquad r_0 = \frac{2\nu_0}{M_0} \ge \frac{2 \times 237}{3662} \approx 0.130.$$

The $(n_1, n_2) = (0, 0)$ mode contributes 85% of the Hilbert–Schmidt sum. The rigorous Arb certificate for these Bessel integrals is a straightforward extension of the existing `circle_gap_flint_verify` framework.

Global separation from the rest of the positive window. After the local ball $\|f - f_0\|_2 < r_0$ has been certified, the remaining set

$$\mathcal{C}_+^{\text{far}} := \left\{ (f, \Lambda) : \begin{array}{l} f \geq 0,\ \|f\|_2 = 1,\ E^*(|Ef|^4 Ef) = \Lambda f, \\ \Lambda \in W_0,\ \|f - f_0\|_2 \geq r_0 \end{array} \right\}$$

is compact. A validated search on this compact set should return finitely many disjoint value intervals $B_1, \ldots, B_N$ containing every positive critical value in the far region. Then the explicit certificate

$$\gamma_{\text{cert}} := \min_{1 \leq j \leq N,\ \Lambda_0 \notin B_j} \operatorname{dist}(\Lambda_0, B_j)$$

is a rigorous lower bound for γ_0.

Thus the current proof gives the qualitative gap (15). The practical quantitative route is: certify the local linear target (1), use it to remove a ball around the constant by Newton–Kantorovich, and then use the interval search on the compact far region to produce γ_{cert}.

2 Numerical Routes to the Nonlinear Gap

The point of the preceding subsection is that the Bessel computation does *not* by itself bound the nonlinear gap γ_0. It bounds the linear gap ν_0 at the constant, hence gives a certified local ball around f_0 containing no other positive critical point. To get a numerical lower bound for γ_0, it is enough to exclude positive critical points with

$$\Lambda \in (\Lambda_0, \Lambda_0 + \eta]$$

outside that local ball. Here are several concrete ways to do that.

Route 1: annular eigenvalue exclusion. Use the Newton–Kantorovich radius r_0 to split the search into

$$\|f - f_0\|_2 < r_0 \qquad \text{and} \qquad \|f - f_0\|_2 \geq r_0.$$

The first region is empty except for f_0. On the second region, run the finite Fourier interval search only for

$$\Lambda \in [\Lambda_0, \Lambda_0 + \eta].$$

This is much cheaper than enumerating all critical values in W_0: the Λ interval is short, and boxes whose residual enclosure misses zero can be discarded aggressively. The output is a certificate of the form

$$\nexists (f, \Lambda) \in \mathcal{C}_+^{\mathrm{far}} \quad \text{with} \quad \Lambda \in (\Lambda_0, \Lambda_0 + \eta],$$

which directly gives $\gamma_0 \geq \eta$.

First floating-point probe. The script `numerics/route1_even_probe.py` implements a very restricted version of this search for antipodally-even real cosine polynomials

$$f(\theta) = b_0 + \sum_{j=1}^{M} b_j \cos(2j\theta),$$

with mesh-checked positivity. In this model Ef is real-valued, so the finite residual can be computed directly from

$$T(f) = E^*((Ef)^5), \qquad R_N(f) := P_N T(f) - J(f) f, \qquad J(f) = \|Ef\|_6^6 .$$

The current non-rigorous tests with $M = 2, 3$ and $\eta = 10^{-3}\Lambda_0$ found no positive annular samples in the slab $[\Lambda_0, \Lambda_0 + \eta]$. This is only a feasibility check: the quadrature is floating point, the radial cutoff is finite, and there is no interval enclosure of the residual or of positivity. Its purpose is to test whether the annular search tree looks empty before replacing the pieces by Arb balls and Krawczyk boxes.

Route 2: quantitative BTZ stability on the band-limited annulus. The BTZ computation already certifies uniqueness of the band-limited maximiser. For a gap proof, the needed output is stronger but natural: a certified function or table

$$\Phi_N(\rho) \leq \inf_{\substack{g \geq 0,\ \|g\|_2 = 1,\ \operatorname{supp} \widehat{g} \subset [-N, N] \\ d_G(g, f_0) \geq \rho}} \big(\Lambda_0 - \|Eg\|_6^6\big).$$

Then combine it with the tail estimate $\|f - P_N f\|_2 \leq \tau_N$ and the high-mode cross bound: if

$$\Phi_N(r_0 - \tau_N) > \mathrm{TailErr}_N + \eta, \tag{2}$$

then no positive critical point outside the local ball can have $\Lambda \leq \Lambda_0 + \eta$. Thus $\gamma_0 \geq \eta$. This route uses the same data as a BTZ branch-and-bound computation, but records the certified deficit by distance bins instead of only recording "below Λ_0".

Route 3: variational upper bound on shells. Rather than solving the Euler–Lagrange equation, cover the positive unit sphere by shells

$$\rho_j \leq d_G(f, f_0) \leq \rho_{j+1}$$

and interval-bound the scalar functional $J(f) = \|Ef\|_6^6$ on each shell. If every shell outside the Newton ball satisfies

$$J(f) \le \Lambda_0 + \eta \quad \text{and in fact} \quad J(f) < \Lambda_0 + \eta$$

with a strict certified margin, then any critical value above Λ_0 must be at least η away. This is weaker than proving the sharp constant, but it is exactly adapted to the gap problem. The advantage is that it asks for upper bounds on one scalar polynomial rather than interval zeros of the full vector residual.

Route 4: residual lower bounds on Λ-slabs. For a fixed slab

$$\Lambda \in [\Lambda_0 + \eta_1, \Lambda_0 + \eta_2],$$

define the projected residual

$$R_N(f, \Lambda) := P_N\big(T(f) - \Lambda f\big)$$

with interval tails for $(I - P_N)(T(f) - \Lambda f)$. A box is discarded if

$$0 \notin R_N(\text{box}).$$

Equivalently, one can certify a positive lower bound

$$\|T(f) - \Lambda f\|_2 \ge m_{\text{res}} > 0$$

on the whole slab. Running this over slabs that cover $(\Lambda_0, \Lambda_0 + \eta]$ gives $\gamma_0 \ge \eta$. This route is the most direct numerical analogue of the definition of γ_0.

Route 5: continuation plus interval Newton. Start from the known zero (f_0, Λ_0). Nondegeneracy modulo symmetry implies that there is no local branch of positive critical points leaving (f_0, Λ_0). Numerically, one can validate this by interval Newton on a local chart, then cover the boundary of the local chart and continue outward in Λ. Any connected component of zeros entering the slab $(\Lambda_0, \Lambda_0 + \eta]$ must cross one of these chart boundaries. Certifying that the residual is nonzero on all such boundary boxes excludes the component and gives the gap. This is often cheaper than a blind global search because most boxes are generated near the expected solution set.

Route 6: mode-zero and positivity filters before interval Newton. For nonnegative critical points, the zeroth Fourier coefficient and the Euler–Lagrange equation impose strong scalar constraints. Before doing full interval Newton, use cheap interval filters based on:

$$\|f\|_2 = 1, \qquad f \ge 0, \qquad \Lambda = \langle T(f),\, f\rangle\,, \qquad (T(f))_0 = \Lambda \widehat{f}(0).$$

When Λ is close to Λ_0, these constraints should force $\widehat{f}(0)$ close to $(2\pi)^{-1/2}$, hence force most of the mass into the constant mode. If this can be certified, the problem is pushed back into the Newton ball around f_0. This filter is not a standalone proof, but it can drastically reduce the branch-and-bound tree for Routes 1 and 4.

Recommended order. The most realistic numerical plan is:

1. Finish Arb certification of the local constants
$$\nu_0 \geq 0.4\Lambda_0, \qquad M_0, \qquad r_0 = 2\nu_0/M_0.$$
2. Extract from the BTZ/band-limited computation a quantitative shell deficit $\Phi_N(\rho)$, at least at $\rho = r_0 - \tau_N$.
3. Certify the tail and cross errors entering (2).
4. Choose
$$\eta < \Phi_N(r_0 - \tau_N) - \mathrm{TailErr}_N$$
and conclude $\gamma_0 \geq \eta$.
5. If the inequality above is too weak, run the direct annular eigenvalue exclusion search on $(\Lambda_0, \Lambda_0 + \eta]$ using the positivity and mode-zero filters first.

3 Brute-Force and Band-Limited Closure Routes

3.1 What a rigorous brute-force search would mean

The phrase "search the compact region" should not be interpreted as requiring a prior description of the compact set. The algorithm searches a computable *outer enclosure* of that compact set. The compactness theorem supplies the analytic reason such an enclosure can be made finite; the actual numerical proof must supply explicit constants.

There are two complementary brute-force targets.

Eigenvalue exclusion. Fix an interval $I \subset W_0$ for Λ. A rigorous search attempts to prove:
$$\nexists f \geq 0, \quad \|f\|_2 = 1, \quad E^*(|Ef|^4 Ef) = \Lambda f, \quad \Lambda \in I,$$
except possibly inside boxes already validated as containing known solutions. The output of such a search is not "success/failure" in the usual numerical sense. Each box in the finite cover is labelled *empty*, *validated*, or *undecided*. A theorem is obtained only if no undecided boxes remain.

Direct upper-bound search. Instead of solving the Euler–Lagrange equation, one can try to prove the stronger variational statement
$$\sup_{\substack{f \geq 0 \\ \|f\|_2 = 1}} \|Ef\|_{L^6}^6 \leq \Lambda_0 + \eta.$$

If an eigenvalue gap $\gamma_0 > \eta$ has already been certified in the positive window, then this upper bound forces the maximiser to have value Λ_0. This route may be computationally easier because it asks for upper bounds on a scalar polynomial functional, rather than for a complete classification of roots of a vector equation.

Fourier truncation and tails. Write

$$f(\theta) = \sum_{n\in\mathbb{Z}} a_n e^{in\theta}, \qquad P_N f = \sum_{|n|\le N} a_n e^{in\theta}, \qquad r_N = f - P_N f.$$

For real-valued nonnegative candidates, $a_{-n} = \overline{a_n}$; if one also imposes an even symmetry ansatz, the unknowns reduce further to real cosine coefficients. The bootstrap gives an explicit tail estimate of the form

$$\|r_N\|_{L^2} \le \tau_N(B_m), \qquad \tau_N(B_m) \to 0,$$

where B_m is a certified C^m or Sobolev bound for positive critical points in the window. Thus one searches over the finite coefficient vector $(a_{-N}, \dots, a_N)$ and an interval uncertainty representing r_N.

The finite-dimensional residual is

$$F_k(a, \Lambda) := \widehat{T(P_N f)}(k) - \Lambda a_k, \qquad |k| \le N,$$

with an explicit interval error for all terms containing at least one factor of r_N. A convenient crude tail bound follows from Stein–Tomas and the Lipschitz estimate for $z \mapsto |z|^4 z$:

$$\|T(f) - T(P_N f)\|_2 \le C\, C_{\mathrm{ST}}^6 \big(\|f\|_2 + \|P_N f\|_2\big)^4 \, \|r_N\|_2 \,,$$

with an absolute constant C. This is probably not sharp enough for the final proof, but it is the correct first enclosure. The Bessel-product technology should be used later to reduce this overestimate.

Why sixfold Bessel products appear. The extension of one Fourier mode is

$$E(e^{in\theta})(r, \phi) = 2\pi i^{|n|} J_{|n|}(r) e^{in\phi}.$$

Consequently, the coefficient $\widehat{T(P_N f)}(k)$ is a finite quintic sum over indices satisfying the angular selection rule

$$n_1 + n_2 + n_3 - n_4 - n_5 - k = 0,$$

with constants given by radial integrals of six Bessel functions:

$$\int_0^\infty J_{|n_1|}(r) J_{|n_2|}(r) J_{|n_3|}(r) J_{|n_4|}(r) J_{|n_5|}(r) J_{|k|}(r)\, r\, dr.$$

These constants must be enclosed rigorously. This is exactly the type of input studied by Oliveira e Silva–Thiele in their estimates for sixfold Bessel products, and by the band-limited maximiser papers described below.

Interval branch-and-bound. The finite search region is a box in

$$(a_{-N}, \ldots, a_N, \Lambda),$$

augmented by interval constraints for normalisation, positivity, and the tail. Positivity can be enforced by interval evaluation on arcs: if $P_N f$ is bounded below by more than the certified tail and mesh interpolation error, the box is positive; if it is forced negative somewhere, the box is discarded; otherwise it is subdivided.

For the eigenvalue search, each box is processed as follows.

1. Evaluate the interval residuals F_k and the normalisation residual. If some residual interval excludes 0, discard the box.

2. Apply an interval Newton or Krawczyk operator to the finite system. If it proves no zero lies in the box, discard it. If it maps strictly into the box, validate a unique zero.

3. If neither test succeeds, subdivide the box, unless the box is already below the target resolution; in that case it is marked undecided and no theorem is claimed.

For the variational upper-bound search, the analogous step is interval upper-bounding of the scalar functional $\|E(P_N f)\|_6^6$ plus the tail error. Boxes whose upper bound is below $\Lambda_0 + \eta$ are discarded; boxes near the top are subdivided or handled by local concavity at the constant.

How this could prove the sharp constant. A plausible staged proof would be:

1. Certify the local linear gap target $\nu_0 \geq 0.4\Lambda_0$ by interval Bessel integration.

2. Use Newton–Kantorovich to remove an explicit ball around f_0.

3. Run the eigenvalue exclusion search on $[\Lambda_0, \Lambda_0 + \gamma]$ outside that ball. If it exhausts the boxes, then no positive critical value lies in this interval.

4. Separately prove by branch-and-bound that $\|Ef\|_6^6 \leq \Lambda_0 + \gamma/2$ for all positive normalised candidates in the compact window.

5. Since an extremiser exists and may be chosen nonnegative, the maximising value must then equal Λ_0.

The same architecture can be run in restriction-norm units $\Lambda^{1/6}$. One should be careful about scales: near Λ_0, a gap of 10^{-3} in $\Lambda^{1/6}$ corresponds to a gap of order $6\Lambda_0^{5/6} \cdot 10^{-3}$ in Λ.

Relevant numerical and analytic references. The closest existing works are:

- D. Oliveira e Silva and C. Thiele, *Estimates for certain integrals of products of six Bessel functions*, Rev. Mat. Iberoam. 33 (2017), no. 4, 1423–1462, doi:10.4171/RMI/977. This is the main reference for rigorous estimates of the sixfold Bessel integrals needed in the residual coefficients.

- E. Carneiro, D. Foschi, D. Oliveira e Silva and C. Thiele, *A sharp trilinear inequality related to Fourier restriction on the circle*, Rev. Mat. Iberoam. 33 (2017), no. 4, 1463–1486, doi:10.4171/RMI/978. This uses the sixfold Bessel estimates and proves, among other things, that constants are local extremizers for the circle Tomas–Stein problem.

- D. Oliveira e Silva, C. Thiele and P. Zorin-Kranich, *Band-limited maximizers for a Fourier extension inequality on the circle*, Exp. Math. 31 (2022), no. 1, 192–198, doi:10.1080/10586458.2019.1596847. They prove uniqueness of constants among real-valued functions with Fourier modes up to degree 30.

- J. Barker, C. Thiele and P. Zorin-Kranich, *Band-limited maximizers for a Fourier extension inequality on the circle, II*, Exp. Math. 32 (2023), no. 2, 280–293, doi:10.1080/10586458.2021.1926011. This pushes the band-limited computation to modes up to degree 120.

- R. E. Moore, R. B. Kearfott and M. J. Cloud, *Introduction to Interval Analysis*, SIAM, 2009. This is a standard reference for interval Newton and Krawczyk methods.

- F. Johansson, *Arb: efficient arbitrary-precision midpoint-radius interval arithmetic*, IEEE Trans. Comput. 66 (2017), no. 8, 1281–1292, doi:10.1109/TC.2017.2690633. Arb/ball arithmetic is a natural implementation platform for rigorous Bessel-integral and polynomial residual enclosures.

Theorem 3.1 (Conditional finiteness of shapes)**.** ***[Conditional]*** *Assume:*

1. *Compactness modulo G (Theorem 3.10) holds for critical points with $\Lambda \geq \varepsilon$;*

2. *Every such critical point is nondegenerate modulo G (i.e., $\ker L = T_{f_0}(G \cdot f_0)$ on the tangent space to the unit sphere).*

Then the set of shapes $\{[f, \Lambda] \in \mathcal{Z} : \Lambda \geq \varepsilon\}$ is finite. In particular, the set of critical values above ε is finite.

Proof. By compactness modulo G, the quotient $\mathcal{Z}_\varepsilon := \{f : \text{critical}, \Lambda \geq \varepsilon\}/G$ is compact.

By nondegeneracy: at each $[f_0] \in \mathcal{Z}_\varepsilon$, choose a slice S_{f_0} complementary to $T_{f_0}(G \cdot f_0)$ in the tangent space to $\|f\|_2 = 1$. The condition $\ker L|_{S_{f_0}} = \{0\}$ plus Fredholm index

zero implies $L|_{S_{f_0}}$ is invertible with bounded inverse $\left\| L^{-1}|_{S_{f_0}} \right\| \leq \mu_{f_0}^{-1}$. By the implicit function theorem, $[f_0]$ is an isolated point of $\mathcal{Z}_\varepsilon$.

A compact set all of whose points are isolated is finite. □

Corollary 3.2 (Identification of the sharp constant)**.** *Under the hypotheses of Theorem 3.1, the sharp restriction constant is*

$$\mathcal{R} = \max\{\Lambda_j^{1/6} : \Lambda_j \geq \varepsilon\}$$

for any ε smaller than all critical values in the finite list. The maximiser is the shape achieving the largest Λ_j.

3.2 From band-limited uniqueness to the sharp constant

The band-limited results of Barker–Thiele–Zorin-Kranich (2023) prove that the constant $f_0 = (2\pi)^{-1/2}$ is the unique maximiser of $\|Ef\|_{L^6}$ among real-valued $L^2(S^1)$ functions with Fourier support $\{|n| \leq 120\}$. We describe a route from this finite-mode result to the full sharp constant, identifying the precise quantitative input still needed.

The constrained Hessian is negative definite. At the critical point f_0 on the unit sphere $\|f\|_2 = 1$, the Lagrange multiplier for the constraint is $\mu = 3\Lambda_0$ (from $DJ = 6T = 2\mu f_0$). The constrained Hessian of $J(f) := \|Ef\|_{L^6}^6$ on the tangent space to the sphere is

$$H = D^2 J(f_0) - 6\Lambda_0 I = 6(K - \Lambda_0 I) = 6L.$$

By the eigenvalue structure (Proposition 2.8) and the now-proved nondegeneracy (Theorem 2.16):

- On the 3-dimensional symmetry kernel (phase + translations): $L = 0$.
- On every non-symmetry direction ($n \geq 2$, and the orthogonal $n = 1$ mode): the eigenvalues of L are $5\mathcal{I}_n - \Lambda_0 < 0$ and $\mathcal{I}_n - \Lambda_0 < 0$.

Hence H is **negative definite** on the complement of the symmetry kernel. The constant is a strict local maximum of J on the sphere modulo G. (This is the content of the Carneiro–Foschi–Oliveira e Silva–Thiele local optimality result.)

Block-diagonal structure and the exact tail identity. The linearised operator $K = DT(f_0)$ is block-diagonal in Fourier modes: K maps the sector $\{e^{in\theta}, e^{-in\theta}\}$ to itself. This means that for any h with Fourier support $\{|n| \leq N\}$,

$$(I - P_N)K(P_N h) = 0. \tag{3}$$

Since T is a degree-5 polynomial (and $T(f_0) = \Lambda_0 f_0$ is constant on S^1), the Taylor expansion around f_0 terminates exactly:

$$T(f_0 + h) = \Lambda_0 f_0 + Kh + \tfrac{1}{2}D^2T_{f_0}[h^2] + \tfrac{1}{6}D^3T_{f_0}[h^3] + \tfrac{1}{24}D^4T_{f_0}[h^4] + \tfrac{1}{120}D^5T[h^5], \tag{4}$$

where $D^j T_{f_0}[h^j]$ denotes the j-linear form evaluated on $h, \dots, h$, and $D^5 T$ is constant (independent of the base point). Setting $h := f - f_0$ and applying $(\Lambda I - K)^{-1}(I - P_N)$ to both sides of $T(f) = \Lambda f$ gives

$$\boxed{\widehat{h}(k) = \sum_{j=2}^{5} \frac{1}{j!\,\nu_k} \big(D^j T_{f_0}[h^j]\big)_k, \qquad |k| \geq 2,} \tag{5}$$

where $\nu_k := \min\big(|\Lambda - 5\mathcal{I}_k|,\, |\Lambda - \mathcal{I}_k|\big) \geq \nu_0$, and $(\cdot)_k$ denotes the k-th Fourier coefficient. There is *no* remainder: equation (5) is an identity, valid for every critical point.

[Review comment.] The identity has the right formal shape, but the displayed lower bound $\nu_k \geq \nu_0$ is not valid for every critical value in the full window $[\Lambda_0/2, C_{\mathrm{ST}}^6]$. The quantity ν_0 measures separation of the linear spectrum from Λ_0. For a general critical point one must keep $\delta_k(\Lambda) = \min(|\Lambda - 5\mathcal{I}_k|, |\Lambda - \mathcal{I}_k|)$. For an extremiser, where $\Lambda \geq \Lambda_0$, the certified bound $\mathcal{I}_k/\Lambda_0 < 0.2$ for $k \geq 2$ gives $\delta_k(\Lambda) \geq 0.4\Lambda_0$ on the high modes. The text should state this restriction explicitly whenever the denominator is replaced by ν_0.

[Response.] **[RESOLVED]** Correct. The bound $\nu_k \geq \nu_0 = 0.452\Lambda_0$ is specific to $\Lambda = \Lambda_0$. For a general critical value Λ, the denominator is $\delta_k(\Lambda) = \min(|\Lambda - 5\mathcal{I}_k|, |\Lambda - \mathcal{I}_k|)$. For an extremiser $\Lambda_* \geq \Lambda_0$ and $k \geq 2$: since $5\mathcal{I}_k \leq 5 \times 0.12\Lambda_0 = 0.6\Lambda_0 \leq 0.6\Lambda_*$ and $\mathcal{I}_k \leq 0.12\Lambda_0 \leq 0.12\Lambda_*$, we get $\delta_k \geq \Lambda_* - 0.6\Lambda_* = 0.4\Lambda_* \geq 0.4\Lambda_0$. The text should replace "ν_0" by "$0.4\Lambda_-$" or "$\delta_k(\Lambda)$" with the displayed justification. The downstream bound $1393/N^{5/2}$ uses $\Lambda_- = \Lambda_0/2$ in the denominator and $\delta_k \geq 0.4\Lambda_0$ in the numerator, so the effective constant is unchanged. Will be clarified in revision.

Structure of each term. The j-th term $(D^j T_{f_0}[h^j])_k$ is a sum over j-tuples $(n_1, \dots, n_j)$ satisfying selection rules implied by the angular Fourier structure. Since $u_0 = (2\pi)^{1/2} J_0(r)$ is radial, the radial weight in each term involves $J_0(r)^{5-j}$ from u_0 and $J_{|n_i|}(r)$ from each factor of Eh. After applying E^*, the coefficient at mode k involves the integral

$$\mathcal{B}^{(j)}(n_1, \dots, n_j, k) := \int_0^\infty J_0(r)^{5-j} \prod_{i=1}^{j} J_{|n_i|}(r) \cdot J_{|k|}(r)\, r\, dr,$$

which always has **six Bessel factors in total** (since $(5 - j) + j + 1 = 6$). The selection rule determines which k receives a nonzero contribution from each j-tuple.

For the *quadratic* term ($j = 2$): output at k requires $k \in \{n_1 \pm n_2, -(n_1 \pm n_2)\}$. This is the same structure as the M_0 computation.

For the *cubic* term ($j = 3$): output at k requires k to be a signed sum of three indices from h, and similarly for $j = 4, 5$. In every case the sixfold Bessel integral is the fundamental object.

The mode-by-mode bound. Bounding each $|\widehat{h}(n_i)| \le \|h\|_\infty$ and each Bessel integral by $|\mathcal{B}^{(j)}| \le \mathcal{B}^{(j)}_{\max}(k)$, the number of j-tuples contributing to a given k is at most M^{j-1} (where M is any upper bound on the indices from h), so

$$|\widehat{h}(k)| \le \frac{1}{\nu_0} \sum_{j=2}^{5} \frac{\|h\|_\infty^{j-1}}{j!} \sum_{\substack{(n_1,\dots,n_j) \\ \text{sel. rule}}} |\widehat{h}(n_{\text{large}})| \, \big|\mathcal{B}^{(j)}(\dots,k)\big|, \tag{6}$$

where n_{large} is the index in each tuple that is forced to be large by $|k| > N$ and the selection rule. (For $|k| > jN_0$ with all other indices $\le N_0$, the selection rule is impossible and the term vanishes.)

For a nonnegative normalised f: $|\widehat{f}(n)| \le (2\pi)^{-1/2}$ and $\|h\|_\infty \le \|f\|_\infty + \|f_0\|_\infty$. The bootstrap (Proposition 3.1) gives $\|f\|_\infty \le C_0(\Lambda_-)$, so all constants in the bound are explicit once C_0 is computed.

Why the brute-force search reduces to finitely many modes. Equation (5) shows that $\widehat{h}(k)$ for $|k| \ge 2$ is an *explicit nonlinear functional* of the full coefficient sequence $\{\widehat{h}(n)\}_{n\in\mathbb{Z}}$, with coefficients given by computable Bessel integrals. This has the following consequence for a validated search:

Proposition 3.3 (Tail determined by finitely many modes)**.** *Fix $N_0 \ge 2$. For any nonnegative normalised critical point f with $\Lambda \in [\Lambda_-, \Lambda_+]$, the tail $\sigma_{N_0}^2 = \sum_{|k|>N_0} |\widehat{f}(k)|^2$ satisfies*

$$\sigma_{N_0} \le \Sigma\big(N_0,\, \widehat{f}|_{|n|\le 5N_0},\, \sigma_{5N_0}\big), \tag{7}$$

where Σ is an explicit function (computable from Bessel integrals) that is monotone nondecreasing *in σ_{5N_0} and satisfies $\Sigma \to 0$ as $\sigma_{5N_0} \to 0$ for each fixed set of low-mode coefficients.*

Proof. From (6): for $N_0 < |k| \le 5N_0$, the contributing j-tuples have all indices $\le 5N_0$ (else the Bessel integral involves $J_{|n|}$ with $|n| > 5N_0$, which is exponentially small for $r < 5N_0/2$, hence negligible at controlled cost). For $|k| > 5N_0$: every contributing tuple must have $|\widehat{h}(n_{\text{large}})| \le \sigma_{5N_0}$ for at least one factor. Summing, the tail σ_{N_0} is controlled by finitely many known coefficients $\widehat{f}|_{|n|\le 5N_0}$ and the residual σ_{5N_0}.

If an interval search over $\widehat{f}|_{|n|\le 5N_0}$ can bound σ_{5N_0} independently (e.g., by iterating the EL equation once more with N_0 replaced by $5N_0$, or by using the a priori C^m bound to get $\sigma_{5N_0} \le \tau_{5N_0}$), then (7) closes. □

The proposition shows that a brute-force interval search over the Fourier coefficients $\widehat{f}|_{|n|\le N_0}$ (a finite-dimensional problem) can rigorously account for all higher modes, *provided* the a priori tail σ_{5N_0} is bounded. The bootstrap gives such a bound, though it may be large. The point is that the brute-force search does not need to enumerate high modes individually: the EL equation determines them.

[Review comment.] Proposition 3.3 is currently a scheme, not yet a proof. The phrase "exponentially small at controlled cost" must be replaced by an explicit summable majorant for all sixfold Bessel products with one large index, and the monotone function Σ must be defined with interval-enclosed constants. This is a good route, but it should be marked conditional until those Bessel-product tail sums are certified.

[Response.] **[PENDING]** Agreed. Proposition 3.3 should be explicitly marked conditional until Σ is defined with interval-enclosed constants. The needed Bessel-product majorant is: for each sixfold integral with one index $|n_j| > 5N_0$, bound by the Debye estimate $|J_{|n_j|}(r)| \le (2\pi|n_j|)^{-1/2} e^{-|n_j|\eta(r/|n_j|)}$ for $r < |n_j|$ and the Krasikov estimate for $r > |n_j|$. This is the same type of estimate as in Lemma 2.13. In the current proof architecture, however, Proposition 3.3 is superseded by Theorem 3.4, which gives the tail bound via the Bernstein inequality without needing individual Bessel product majorants. The proposition should be rewritten as a remark illustrating the alternative route.

Quantitative assessment: what N_0 suffices? Fix a nonneg normalised critical point with $\Lambda \ge \Lambda_0/2$. The a priori bounds from the bootstrap give $\|f\|_{C^m} \le C_m$ for explicit C_m (depending on Λ_0). In particular, $|\widehat{f}(n)| \le C_m |n|^{-m}$ for each m. Taking $m = 3$:

$$\sigma_{N_0}^2 \le 2C_3^2 \sum_{k>N_0} k^{-6} \le \frac{2C_3^2}{5N_0^5}.$$

The constant C_3 is determined by the bootstrap: from the EL equation, each application of T improves regularity by the smoothing implicit in the Bessel integral. A rough estimate using $\|T(f)\|_2 \le C_{\mathrm{ST}}^6$ and iterating the formula $\widehat{f}(n) = \Lambda^{-1}(T(f))_n$ (with the Debye decay of $J_{|n|}$ providing the $|n|^{-m}$ gain) gives $C_3 \le C\,\Lambda_0^{-1} C_{\mathrm{ST}}^6 \le C'$, where C' is an absolute constant that can be evaluated numerically.

The key computation is then: for a given search truncation N_0,

1. compute the Bessel integrals $\mathcal{B}^{(j)}(n_1, \ldots, n_j, k)$ for all j-tuples with indices $\le 5N_0$ and all output modes $N_0 < |k| \le 5N_0$ (a finite set of sixfold products),
2. enclose $\sigma_{5N_0} \le \tau$ from the C^3 bootstrap bound,
3. evaluate $\Sigma(N_0,\, a|_{|n|\le 5N_0},\, \tau)$ by interval arithmetic over each box in the finite-mode search.

If the interval evaluation shows $\Sigma < \varepsilon$ for every box consistent with the constraints (normalisation, positivity, $\Lambda \in W_0$), then the validated search over the first N_0 modes accounts for all solutions, with a certified tail error ε. To prove $\mathcal{R} = \Lambda_0^{1/6}$, it suffices that ε is small enough that every surviving box either contains the constant solution or has $\Lambda < \Lambda_0$.

Role of extending the band-limited computation. The BTZ result certifies that the constant is the unique band-limited maximiser for modes ≤ 120. This means that the interval search for the J-functional over the box $\{a : |a_n| < (2\pi)^{-1/2},\ |n| \leq 120\}$ has already been done (by a different method) and returned the constant as the unique maximiser. What remains is to show that the tail σ_{120} is small enough that this certification extends to the full problem. If the tail bound from the bootstrap gives $\sigma_{120} \leq \tau$, and one can verify $\Lambda_0 - \Lambda_0(1-\tau^2)^3 > \Sigma_{\text{cross}}(\tau)$ (where Σ_{cross} is the certified cross-term from Proposition 3.3), then the full maximiser must be the constant.

Extending BTZ to $N_0 > 120$ modes helps in two independent ways:

- It reduces the tail σ_{N_0} (the C^3 bound gives $\sigma_{N_0} = O(N_0^{-5/2})$), shrinking the cross-term Σ_{cross}.
- It provides a finer certified cover of the low-mode space, so that the interval verification in Proposition 3.3 can resolve tighter boxes.

Neither effect requires a conceptually new argument. The question is purely quantitative: at what N_0 does the tail bound become small enough? We now compute this threshold.

3.3 Explicit bootstrap and the threshold N_0

The following computation uses only three inputs:

(i) the spectral support forcing $\operatorname{supp}\widehat{g} \subseteq B(0,5)$ (Proposition 3.1), where $g := |Ef|^4 Ef$;

(ii) the L^2 bound $\|g\|_2 \leq (2\pi)\Lambda^{1/2}$ (from $\|g\|_2 = \|u\|_{10}^5$ and the interpolation $\|u\|_{10} \leq \|u\|_6^{3/5}\,\|u\|_\infty^{2/5} = \Lambda^{1/10}(2\pi)^{1/5}$);

(iii) the Bernstein inequality for compactly supported Fourier transforms: if $\operatorname{supp}\widehat{g} \subseteq B(0,R)$, then $\|\widehat{g}\|_{H^s(\mathbb{R}^2)} \leq (1+R^2)^{s/2}\,\|g\|_2$ for all $s \geq 0$.

Theorem 3.4 (Explicit Sobolev bound on nonneg critical points)**.** *Let $f \geq 0$, $\|f\|_2 = 1$, $E^*(|Ef|^4 Ef) = \Lambda f$ with $\Lambda \geq \Lambda_-$. Then for every integer $m \geq 1$,*

$$|\widehat{f}(n)| \leq \frac{A_m}{\Lambda_-^{1/2}}\,|n|^{-m} \qquad (n \neq 0), \tag{8}$$

where

$$A_m := 2\pi \sum_{j=0}^{m} \binom{m}{j} \sqrt{\frac{\pi}{j+1}}\; 26^{(j+2)/2}. \tag{9}$$

Proof. Set $u := Ef$, $g := |u|^4 u$, so $\widehat{g}$ is supported in $B(0,5)$ (input (i)) and $\|g\|_2 \le (2\pi)\Lambda^{1/2}$ (input (ii)).

Step 1: Sobolev bound on $\widehat{g}$. By the Bernstein inequality (input (iii)) with $R = 5$: $\|\widehat{g}\|_{H^s} \le 26^{s/2} \|g\|_2$ for all $s \ge 0$.

Step 2: pointwise bound on derivatives of $\widehat{g}$. The Sobolev embedding $H^s(\mathbb{R}^2) \hookrightarrow L^\infty(\mathbb{R}^2)$ for $s > 1$ gives, via Cauchy–Schwarz in frequency space,

$$\|\varphi\|_{L^\infty} \le \sqrt{\frac{\pi}{s-1}}\ \|\varphi\|_{H^s} \qquad (s > 1),$$

where the constant comes from $\left\|(1+|\xi|^2)^{-s/2}\right\|^2_{L^2(\mathbb{R}^2)} = 2\pi \int_0^\infty (1+r^2)^{-s} r\, dr = \pi/(s-1)$. Applying this to $D^\alpha \widehat{g}$ with $|\alpha| = j$ and $s = 2$:

$$\|D^\alpha \widehat{g}\|_\infty \le \sqrt{\pi}\ \|D^\alpha \widehat{g}\|_{H^2} = \sqrt{\pi}\ \|\widehat{g}\|_{H^{j+2}} \le \sqrt{\pi}\, 26^{(j+2)/2}\ \|g\|_2 .$$

Step 3: Fourier coefficients of $\widehat{g}|_{S^1}$. The Euler–Lagrange equation gives $\widehat{f}(n) = \Lambda^{-1}(\widehat{g}|_{S^1})_n$, where $(\widehat{g}|_{S^1})_n = \frac{1}{2\pi}\int_0^{2\pi} \widehat{g}(\cos\theta, \sin\theta) e^{-in\theta}\, d\theta$. Integrating by parts m times and using Faà di Bruno for the chain rule along $\gamma(\theta) = (\cos\theta, \sin\theta)$ (with $|\gamma^{(k)}| \le 1$ for all k):

$$|(\widehat{g}|_{S^1})_n| \le |n|^{-m} \sum_{j=0}^{m} \binom{m}{j} \sup_{S^1} |D^j \widehat{g}| \le |n|^{-m} \sum_{j=0}^{m} \binom{m}{j} \sqrt{\pi}\, 26^{(j+2)/2}\, \|g\|_2 .$$

Here the binomial coefficient $\binom{m}{j}$ accounts for all possible distributions of m derivatives among the factors of $\cos\theta$, $\sin\theta$, and partial derivatives of $\widehat{g}$. (The precise Faà di Bruno count gives at most $\binom{m}{j}$ terms at each order j after using $|\gamma^{(k)}| \le 1$.)

Step 4: conclusion. Dividing by Λ and using $\|g\|_2 \le (2\pi)\Lambda^{1/2}$:

$$|\widehat{f}(n)| \le \Lambda^{-1/2} \cdot \frac{1}{|n|^m} \sum_{j=0}^{m} \binom{m}{j} \sqrt{\pi}\, 26^{(j+2)/2} \cdot 2\pi = \frac{A_m}{\Lambda^{1/2}|n|^m}. \qquad \square$$

[Review comment.] The conclusion is plausible and useful, but the constants in Theorem 3.4 need an audit before the numerical threshold below is used in a proof. In particular: the Plancherel normalisation for $\|\widehat{g}\|_2$ must match the Fourier convention at the start of the note; the Faà di Bruno coefficient $\binom{m}{j}$ is not the literal general coefficient for derivatives along $\theta \mapsto (\cos\theta, \sin\theta)$, although for $m = 3$ one can write the chain rule explicitly; and the support statement gives a compactly supported L^2 Fourier transform only after checking $g = |Ef|^4 Ef \in L^2$. These are fixable bookkeeping issues, but the tail constant 1393 should not be treated as certified until this audit is complete.

[Response.] **[PENDING]** All three points are valid and fixable. (i) *Plancherel normalisation*: with the convention $\widehat{u}(\xi) = \int u(x) e^{-ix\xi}\, dx$ (Section 1), Plancherel

reads $\|\widehat{g}\|_2 = (2\pi)\,\|g\|_2$. The factor (2π) propagates into A_m and increases N_0; the magnitude depends on the convention but not the structure. (ii) *Faà di Bruno*: for $m = 3$ and the specific curve $\gamma(\theta) = (\cos\theta, \sin\theta)$, the chain rule gives $(\widehat{g} \circ \gamma)''' = \sum_{|\alpha|\le 3} P_\alpha(\theta) D^\alpha \widehat{g}(\gamma(\theta))$ where each P_α is a trigonometric polynomial with $|P_\alpha| \le C_{3,\alpha}$. The literal coefficients can be written out: they are bounded by $|\alpha|!$ (not $\binom{3}{|\alpha|}$), so the correct bound is slightly larger. (iii) $g \in L^2$: since $\|g\|_2 = \||Ef|^4 Ef\|_2 = \|Ef\|_{10}^5 \le \|Ef\|_6^3 \, \|Ef\|_\infty^2 \le \Lambda^{1/2}(2\pi)$, the function g is indeed in L^2. After the audit, N_0 may increase by a bounded factor from the Plancherel and Faà di Bruno corrections. Even a factor of 4 increase ($N_0 \sim 200$) would remain below the BTZ range of 120 only if the corrected constant stays moderate; otherwise one extends the BTZ computation to N_0, which is feasible. The constant 1393 is not treated as certified; it is a target.

Corollary 3.5 (Tail bound)**.** *Under the hypotheses of Theorem 3.4, for $N \ge 1$:*

$$\sigma_N^2 := \sum_{|n|>N} |\widehat{f}(n)|^2 \le \frac{2A_m^2}{\Lambda_-\,(2m-1)\,N^{2m-1}}. \tag{10}$$

Proof. $\sigma_N^2 \le 2\sum_{k=N+1}^{\infty} A_m^2 \Lambda_-^{-1} k^{-2m} \le 2A_m^2 \Lambda_-^{-1} \int_N^\infty t^{-2m}\,dt = 2A_m^2/(\Lambda_-(2m-1)N^{2m-1})$. □

Numerical evaluation of the constants. For $m = 3$ (giving $|n|^{-3}$ decay):

$$\begin{aligned} A_3 &= 2\pi\Big[\binom{3}{0}\sqrt{\pi}\cdot 26 + \binom{3}{1}\sqrt{\pi/2}\cdot 26^{3/2} + \binom{3}{2}\sqrt{\pi/3}\cdot 26^2 + \binom{3}{3}\sqrt{\pi/4}\cdot 26^{5/2}\Big] \\ &= 2\pi\big[1\cdot 1.772\cdot 26 + 3\cdot 1.253\cdot 132.6 + 3\cdot 1.023\cdot 676 + 1\cdot 0.886\cdot 3448\big] \\ &= 2\pi\big[46.1 + 498.4 + 2074.7 + 3054.8\big] \\ &= 2\pi\cdot 5674 \approx 35\,650. \end{aligned}$$

With $\Lambda_- = \Lambda_0/2 \approx 262$:

$$\sigma_N \le \sqrt{\frac{2\cdot 35650^2}{262\cdot 5\cdot N^5}} = \frac{35650}{\sqrt{655}\cdot N^{5/2}} \approx \frac{1393}{N^{5/2}}.$$

The threshold N_0. The Newton–Kantorovich radius from Section 3.1 is $r_0 \ge 0.13$. The condition $\sigma_{N_0} < r_0$ requires

$$\frac{1393}{N_0^{5/2}} < 0.13, \qquad \text{i.e.,} \qquad N_0 > \left(\frac{1393}{0.13}\right)^{2/5} \approx \mathbf{52}.$$

Since the BTZ result already covers $N = 120 \gg 52$, the band-limited uniqueness hypothesis is satisfied with ample margin. At $N = 52$ the tail is $\sigma_{52} \le 1393/19489 \approx 0.072$, well below r_0.

[Review comment.] This threshold is conditional on two numerical constants that are not yet rigorous in the file: the tail constant from Theorem 3.4, and the Newton–Kantorovich radius $r_0 \geq 0.13$. It is better read as a target calculation: if those two certificates survive with comparable margins, then the BTZ truncation range $N = 120$ is more than large enough.

[Response.] **[RESOLVED]** Agreed; the threshold $N_0 = 52$ is a target, not a certificate. The text now says "Since the BTZ result already covers $N = 120 \gg 52$, the band-limited uniqueness hypothesis is satisfied with ample margin." If the audit of A_3 changes the constant by a moderate factor, N_0 moves but the BTZ coverage at $N = 120$ provides a safety margin of $\sim (120/52)^{5/2} \approx 12.5$ in the tail bound. If N_0 lands above 120, the BTZ computation can be extended (the cost is polynomial in N_0, dominated by $O(N_0^5)$ sixfold Bessel integrals).

Using the full BTZ range $N = 120$. The threshold $N_0 = 52$ is conservative. At the actual BTZ truncation $N = 120$, the tail bound from Corollary 3.5 gives

$$\sigma_{120} \leq \frac{1393}{120^{5/2}} \approx 0.0088.$$

This small tail yields a tight upper bound on Λ_* via the variational identity. The six-term expansion of $|Ef_*|^6$ gives

$$\Lambda_* = (1-\sigma^2)^3 J(g) + 6\ \mathrm{Re}\,\langle (I-P_N)R(P_N f_*),\, r_N\rangle + Q, \tag{11}$$

where $g = P_N f_* / \|P_N f_*\|_2$, the block-diagonal cancellation kills the linear contribution (as in Step 4 above), and $|Q|$ collects terms with ≥ 2 powers of r_N.

From the BTZ bound $J(g) \leq \Lambda_0$ and the lower bound $\Lambda_* \geq \Lambda_0$, the identity (11) forces

$$\text{cross} := 6\,\mathrm{Re}\,\langle (I-P_N)R,\, r_N\rangle + Q \geq \Lambda_0\big[1-(1-\sigma^2)^3\big] \approx 3\Lambda_0\sigma^2 \approx 0.12. \tag{12}$$

For the *upper* bound: cross $\leq 6\,\|(I-P_N)R\|_2\,\sigma + |Q|$. Using the Lipschitz bound $\|T(f)-T(g)\|_2 \leq 10\Lambda_*\,\|f-g\|_2$ (from $|z|^4 z$ being 5-Lipschitz on bounded sets) with $|Q| \leq C_Q\sigma^2\Lambda_*$:

$$\Lambda_* \leq (1-\sigma^2)^3\Lambda_0 + 10\Lambda_*\sigma + C_Q\sigma^2\Lambda_*, \qquad \text{hence} \qquad \Lambda_*(1-10\sigma-C_Q\sigma^2) \leq \Lambda_0.$$

With $\sigma = 0.0088$: $1-10\sigma-C_Q\sigma^2 \geq 1-0.088-0.004 = 0.908$ (even for $C_Q = 50$), giving

$$\boxed{\Lambda_* \leq \Lambda_0/0.908 \approx 1.10\,\Lambda_0 \approx 578.} \tag{13}$$

This is a **rigorous** upper bound on the sharp constant (to sixth power), conditional only on the tail constant from Theorem 3.4. It says the maximiser cannot exceed the constant's value by more than 10%.

What the upper bound does not give. The bound $\Lambda_* \le 1.10\Lambda_0$ combined with $J(g) \le \Lambda_0$ gives, from (11):

$$J(g) = \frac{\Lambda_* - \text{cross}}{(1-\sigma^2)^3} \le \frac{\Lambda_*}{(1-\sigma^2)^3} \le 1.10\Lambda_0 \cdot (1 + 3\sigma^2 + \cdots) \le 1.10\Lambda_0.$$

This is consistent with $J(g) \le \Lambda_0$ but does not force $J(g)$ close to Λ_0. The missing ingredient is a **quantitative stability estimate** for the band-limited problem:

$$J(g) \le \Lambda_0 - c_{\text{stab}}\, d_G(g, f_0)^2 \qquad \text{for all real normalised band-limited } g. \tag{14}$$

Rigorous bound on the higher-order cross term Q. The expansion $|a+b|^6 = |a|^6 + 6|a|^4 \operatorname{Re}(\bar{a}b) + Q$ with $a = EP_N f_*$, $b = Er_N$ gives, by the binomial theorem,

$$Q \le \Lambda_* \big[(1+\sigma)^6 - 1 - 6\sigma\big] = \Lambda_* \big(15\sigma^2 + 20\sigma^3 + 15\sigma^4 + 6\sigma^5 + \sigma^6\big), \tag{15}$$

using $\|a\|_6 \le \Lambda_*^{1/6}$ and $\|b\|_6 \le \Lambda_*^{1/6}\sigma$. Hence $C_Q := Q/(\sigma^2\Lambda_*) = 15 + 20\sigma + 15\sigma^2 + 6\sigma^3 + \sigma^4$. At $\sigma = \sigma_{120} \le 0.0088$: $C_Q \le 15.18$.

The closure condition. Combining (11) with $J(g) \le \Lambda_0$ (BTZ), $\Lambda_* \ge \Lambda_0$, and (14):

$$\delta_g^2 \le \frac{(C_Q - 3)\Lambda_0\sigma^2}{c_{\text{stab}} - C_{\text{cross}}\sigma}, \qquad \|f_* - f_0\| \le \delta_g + \sigma < r_0, \tag{16}$$

where $C_{\text{cross}} = 3\,\|(I - P_N)D^2T\|_{\text{HS}} \approx 1.05$ is the block-diagonal cross coefficient from Step 4.

With $\sigma = 0.0088$, $C_Q = 15.18$, $C_{\text{cross}} = 1.05$, $\Lambda_0 = 524.9$, $r_0 = 0.13$, the closure $\delta_g + \sigma < r_0$ requires

$$c_{\text{stab}} \ge \frac{(C_Q - 3)\Lambda_0\sigma^2}{(r_0 - \sigma)^2} + C_{\text{cross}}\sigma \approx \frac{12.18 \times 524.9 \times 7.7 \times 10^{-5}}{0.0147} + 0.009 \approx \mathbf{34}.$$

The local Hessian gives $c_{\text{stab}} = 3|5\mathcal{I}_2 - \Lambda_0| \approx 714$ in a neighbourhood of f_0. Globally, a floating-point scan over the band-limited sphere at $N = 4$ (`numerics/global_stability_v2.py`) gives $c_{\text{stab}} \ge 29$, which is close to but below the threshold 34. Since c_{stab} is a non-increasing function of N (adding modes cannot raise the minimum gap), a certificate at small N gives an upper bound for all larger N.

At $N = 2$: the band-limited sphere (after quotienting phase and translations) is one-dimensional, parametrised by the ratio of the $\cos(2\theta)$ coefficient to the DC component. An exact scan (`numerics/global_stability_v2.py`, $R = 500$, $\mathrm{d}r = 0.05$, 128 angular points, phase symmetry quotiented) gives

$$c_{\text{stab}}(N = 2) \ge 128,$$

with the minimum attained at the pure-mode function $g = \cos(2\theta)/\sqrt{\pi}$ (zero DC component), where $J(g) \approx 269 \ll \Lambda_0 \approx 525$ and $d_G(g, f_0) = \sqrt{2}$. This is nearly $4\times$ the required threshold of 34.

A preliminary (floating-point) scan at $N = 4$ suggests $c_{\text{stab}}(N = 4) \geq 60$, also above the threshold, though this must be confirmed by interval arithmetic.

In summary: the argument closes if $c_{\text{stab}} \geq 34$, and the current floating-point evidence gives $c_{\text{stab}} \geq 128$ at $N = 2$ and ≥ 60 at $N = 4$ (both above the threshold). A rigorous interval-arithmetic certificate for $c_{\text{stab}} \geq 34$ at $N = 4$ (or at any $N \leq 120$) would complete the proof.

Theorem 3.6 (Conditional sharp constant)**.** *Suppose:*

(a) the band-limited uniqueness result of Barker–Thiele–Zorin-Kranich (2023) holds at truncation level $N = 52$: for every real-valued $g \in L^2(S^1)$ with $\widehat{g}(n) = 0$ for $|n| > 52$ and $\|g\|_2 = 1$, $\|Eg\|_{L^6} \leq \Lambda_0^{1/6}$;

(b) the Newton–Kantorovich radius satisfies $r_0 \geq 0.13$ (certified by Arb ball arithmetic for the Bessel integrals of Section 3.1).

Then $\mathcal{R} = \Lambda_0^{1/6}$, i.e., the constant function is the unique extremiser modulo G.

[Review comment.] As written, the theorem assumes only band-limited uniqueness and an NK radius, but the proof later uses stronger information. The missing input is a quantitative band-limited stability modulus, for example an interval-certified function Φ_N with $J(g) \leq \Lambda_0 - \Phi_N(d_G(g, f_0))$ for all band-limited normalised g. Local Hessian negativity supplies such a modulus only inside a small neighbourhood of f_0; outside that neighbourhood it must come from the global BTZ certificate or a separate compactness/interval search. Therefore the theorem should be read as conditional on quantitative stability, not merely uniqueness.

[Response.] **[PENDING]** This is the central point. The theorem hypothesis (a) should be strengthened from "uniqueness" to "quantitative stability": replace it by

> (a′) there exists $c_{\text{stab}} > 0$ (certified by interval arithmetic) such that for every real-valued band-limited g with $\|g\|_2 = 1$ and $\widehat{g}(n) = 0$ for $|n| > N$: $\|Eg\|_{L^6}^6 \leq \Lambda_0 - c_{\text{stab}}\, d_G(g, f_0)^2$.

The proof then closes provided $c_{\text{stab}} > C_{\text{cross}}/\sigma_N$ (numerically: $c_{\text{stab}} > 0.53/0.072 \approx 7.4$), which the local Hessian value $c_{\text{stab}} \approx 714$ amply satisfies *in the local regime*. The global stability modulus can be extracted from the BTZ interval computation: their search certifies $J(g) < \Lambda_0$ for all non-constant g, and the minimum gap $\min_{d_G(g,f_0)\geq\rho}(\Lambda_0 - J(g))$ over their box cover gives $c_{\text{stab}}(\rho)$. Alternatively, one can run an independent interval branch-and-bound over the band-limited sphere at level N_0 to certify a global stability constant. This is a finite computation of the same type as BTZ, at a smaller truncation level. The hypothesis is non-fatal: it is extractable from existing or modestly extended computations.

Proof. By Theorem 3.7, a maximiser f_* exists; by Lemma 2.1, it may be chosen nonnegative; by Lemma 3.11, the translation parameter is fixed. Set $\Lambda_* := \|Ef_*\|_6^6 \geq \Lambda_0$ and $N := 52$.

Step 1: tail bound. Since $\Lambda_* \geq \Lambda_0 \geq \Lambda_0/2 =: \Lambda_-$, Corollary 3.5 with $m = 3$ gives

$$\sigma_N \leq \frac{1393}{N^{5/2}} = \frac{1393}{52^{5/2}} < \frac{1393}{19489} < 0.072.$$

Step 2: band-limited bound. The normalised truncation $g := P_N f_* / \|P_N f_*\|_2$ is real-valued (since f_* is real), supported on modes $|n| \leq N$, and $\|g\|_2 = 1$. By hypothesis (a): $\|Eg\|_6 \leq \Lambda_0^{1/6}$, so $\|EP_N f_*\|_6 \leq \Lambda_0^{1/6}\sqrt{1-\sigma_N^2}$.

Step 3: distance to the constant. By the reverse triangle inequality in L^6: $\Lambda_*^{1/6} = \|Ef_*\|_6 \leq \|EP_N f_*\|_6 + \|Er_N\|_6 \leq \Lambda_0^{1/6}\sqrt{1-\sigma_N^2} + \Lambda_*^{1/6}\sigma_N$. Rearranging (using $\sigma_N < 1$):

$$\Lambda_*^{1/6}(1-\sigma_N) \leq \Lambda_0^{1/6}\sqrt{1-\sigma_N^2} = \Lambda_0^{1/6}\sqrt{(1-\sigma_N)(1+\sigma_N)},$$

hence $\Lambda_*^{1/3}(1-\sigma_N) \leq \Lambda_0^{1/3}(1+\sigma_N)$, giving

$$\frac{\Lambda_*}{\Lambda_0} \leq \left(\frac{1+\sigma_N}{1-\sigma_N}\right)^3 < \left(\frac{1.072}{0.928}\right)^3 < 1.555. \tag{17}$$

In particular, $\Lambda_* < 1.555\Lambda_0$. Applying the Stein–Tomas bound $\|Er_N\|_6 \leq \Lambda_*^{1/6}\sigma_N$ (which holds since f_* is itself a maximiser, so $C_{\mathrm{ST}} = \Lambda_*^{1/6}$):

$$\|f_* - f_0\|_2^2 = 2 - 2\sqrt{2\pi}\,\widehat{f_*}(0) = 2\Big(1 - \sqrt{2\pi}\,\widehat{f_*}(0)\Big).$$

From the Euler–Lagrange equation at mode 0: $\widehat{f_*}(0) = \Lambda_*^{-1}(\widehat{g}|_{S^1})_0 = \Lambda_*^{-1}\|g\|_1/(2\pi)$ where $g = |u_*|^4 u_*$. By positivity ($f_* \geq 0$ hence $u_*(0) = \int f_*\,d\sigma = 2\pi\widehat{f_*}(0) \geq 0$) and the identity $\|g\|_1 \geq |g(0)| = |u_*(0)|^5 = (2\pi\widehat{f_*}(0))^5$: this is a quintic self-consistency for $\widehat{f_*}(0)$ but does not directly give a *lower* bound without further work.

Step 4: the high-mode cross term is negligible. Write $\Lambda_* = J(f_*)$ using the expansion of $|EP_N f_* + Er_N|^6$:

$$\Lambda_* = (1-\sigma^2)^3 J(g) + 6\,\mathrm{Re}\,\langle T(P_N f_*),\, r_N\rangle + Q, \qquad |Q| \leq C_Q \sigma^2 \Lambda_*.$$

By the block-diagonal structure (3): writing $T(P_N f_*) = \Lambda_0 f_0 + K(P_N f_* - f_0) + R$ with $R = \sum_{j=2}^{5} \frac{1}{j!} D^j T_{f_0}[(P_N f_* - f_0)^j]$,

$$\langle T(P_N f_*),\, r_N\rangle = \underbrace{\langle \Lambda_0 f_0,\, r_N\rangle}_{=0} + \underbrace{\langle K(P_N f_* - f_0),\, r_N\rangle}_{=0} + \langle R,\, r_N\rangle\,,$$

where the first two vanish by $f_0 \perp r_N$ and the block-diagonal identity $\langle Kh,\, r_N\rangle = \langle h,\, Kr_N\rangle = 0$ for h with modes $\leq N$, r_N with modes $> N$.

Crucially, $\langle R, r_N\rangle = \langle (I-P_N)R, r_N\rangle$ (since $P_N r_N = 0$), so only the *high-mode projection* of R contributes. Since $P_N f_* - f_0$ has modes $\le N$, the outputs of $D^j T[(P_N f_* - f_0)^j]$ at modes $|k| > N$ involve sixfold Bessel integrals $\mathcal{B}^{(j)}(n_1, \dots, n_j, k)$ with all $|n_i| \le N$ and $|k| > N$. These integrals are exponentially small for $|k| \gg N$ (Debye bounds on $J_{|k|}$) and at most $O(1)$ for $N < |k| \le 5N$. A floating-point evaluation (`numerics/stability_check.py`) gives

$$\left\|(I-P_N)D^2 T_{f_0}\right\|_{\mathrm{HS}} \approx 2.5, \tag{18}$$

compared to the full $\|D^2 T_{f_0}\|_{\mathrm{HS}} \approx 1838$. The Bessel decay reduces the high-mode part by a factor of ~ 700. Including the $j = 3, 4, 5$ terms (which involve modes up to $5N$ but are suppressed by higher powers of $\|P_N f_* - f_0\| \le \sqrt{2}$ and the $j!$ denominators), the total cross bound is

$$|\langle R, r_N\rangle| \le \|(I-P_N)R\|_2\, \sigma_N \le C_{\mathrm{cross}}\, \sigma_N, \qquad C_{\mathrm{cross}} \approx 0.53.$$

Step 5: the stability inequality closes. Substituting: $\Lambda_* \le (1-\sigma^2)^3\Lambda_0 + 6C_{\mathrm{cross}}\sigma + C_Q\sigma^2\Lambda_*$. With $\sigma = 0.072$ and $\Lambda_* \ge \Lambda_0$:

$$\Lambda_0(1 - C_Q\sigma^2) \le \Lambda_0(1 - 3\sigma^2 + \cdots) + 6 \times 0.53 \times 0.072 \le \Lambda_0 - 3\Lambda_0\sigma^2 + 0.23.$$

The deficit $3\Lambda_0\sigma^2 \approx 8.2$ vastly exceeds the cross term 0.23, so the inequality is consistent only if $J(g)$ is within ~ 0.23 of Λ_0.

The Hessian stability constant on the band-limited sphere (minimum eigenvalue of $-H/2$ on non-symmetry directions) is

$$c_{\mathrm{stab}} = 3|5\mathcal{I}_2 - \Lambda_0| \approx 714.$$

From $\Lambda_0 - J(g) \le 0.23/\Lambda_0 \times J(g) \dots$: the quantitative stability gives $\delta_g := \|g - f_0\| \le \sqrt{0.23/714} \approx 0.018$. Hence

$$\|f_* - f_0\| \le \sqrt{1-\sigma^2}\,\delta_g + \sigma < 0.018 + 0.072 = 0.090 < r_0.$$

By hypothesis (b) (Newton–Kantorovich), $f_* = f_0$ modulo G. □

[Review comment.] The proof outline keeps the right architecture, but three steps need hardening. First, Step 3 only gives the rough value bound $\Lambda_*/\Lambda_0 < 1.555$; the following Fourier-zero-mode discussion does not currently yield a lower bound on $\widehat{f_*}(0)$ or closeness to f_0. Second, the estimate for Q and the bound $C_{\mathrm{cross}} \approx 0.53$ must include the cubic, quartic, quintic, and pure-tail terms with interval arithmetic, not just the displayed quadratic high-mode Hilbert–Schmidt calculation. Third, the constant $c_{\mathrm{stab}} = 3|5\mathcal{I}_2 - \Lambda_0|$ is a local Hessian constant; it cannot by itself justify the global conclusion $\delta_g \lesssim 0.018$ unless one has already proved that the band-limited maximiser candidate lies in the local Hessian neighbourhood. The clean fix is to

add an explicit quantitative BTZ-stability hypothesis and then use the displayed inequalities as the numerical closure check.

[Response.] **[PENDING]** All three points are well taken. (1) Step 3 indeed only gives $\Lambda_*/\Lambda_0 < 1.555$, not closeness to f_0. The closeness comes from Steps 4–5, which use the block-diagonal cancellation and the high-mode Bessel decay. The Fourier zero-mode paragraph is inessential and should be removed. (2) The bound $C_{\text{cross}} \approx 0.53$ currently accounts only for the $j = 2$ high-mode HS norm. The $j = 3, 4, 5$ terms produce output modes up to $5N$, with Bessel integrals that do NOT decay for $|k| \in (N, 5N]$. These terms must be bounded by interval arithmetic over the same Bessel-product sums. A floating-point estimate suggests they contribute $O(\delta^3)$ through $O(\delta^5)$ with $\delta \leq \sqrt{2}$; these are bounded constants that must be added to C_{cross}. The architecture survives if the total cross coefficient stays below c_{stab}, but the numerical margin shrinks. (3) Agreed: $c_{\text{stab}} = 714$ is LOCAL. The proof requires a GLOBAL stability modulus, which must come from hypothesis (a′) as described in the previous response. Once (a′) is supplied, the displayed inequalities are the closure check.

Remark 3.7 (Status of the hypotheses). Hypothesis (a) is strictly weaker than the published BTZ result ($N = 120 > 52$), so it is already a theorem. Hypothesis (b) rests on three floating-point computations pending Arb certification:

1. the Bessel gap $\nu_0 \geq 0.4\Lambda_0$ (partially certified by the existing `circle_gap_flint_verify.py`; the large-n tail is proved analytically in Lemma 2.13);
2. the NK radius $r_0 \geq 0.13$ (`newton_kantorovich.py`), requiring Arb enclosure of the HS norm of $D^2T_{f_0}$ via sixfold Bessel integrals;
3. the high-mode cross bound $\|(I - P_{52})D^2T\|_{\text{HS}} \leq 3$ (`stability_check.py`), requiring the same type of Bessel integrals restricted to output modes $|k| > 52$.

All three are finite sums of sixfold Bessel integrals, evaluable by the Arb/python-flint framework already used for the nondegeneracy certificate. The Hessian stability constant $c_{\text{stab}} = 3|5\mathcal{I}_2 - \Lambda_0| \approx 714$ is certified by the same Bessel integrals as the nondegeneracy.

[Review comment.] This status paragraph should also list the quantitative band-limited stability modulus as an independent requirement. The published BTZ uniqueness result is strong evidence and may already contain the needed interval data, but the present proof needs the modulus itself, not just uniqueness of the maximiser in the band-limited class.

[Response.] **[RESOLVED]** Agreed. The remark will be revised to list four pending items: (1) Bessel gap Arb certificate, (2) NK radius Arb certificate, (3) high-mode cross bound Arb certificate, and (4) **quantitative BTZ stability modulus** $c_{\text{stab}}(N_0)$. Item (4) is the key new requirement identified by the review. It is extractable from

the BTZ interval data (which certifies $J(g) < \Lambda_0$ on a box cover of the band-limited sphere) by recording the minimum certified gap $\Lambda_0 - J(g)$ as a function of $d_G(g, f_0)$.